



UNIVERSITY OF RAJSHAHI

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSE4261 NEURAL NETWORKS AND DEEP LEARNING

Assignment-3

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Contents

1	Introduction	2
2	Methodology	2
2.1	Model and Datasets	2
2.2	Feature Extraction Process	2
2.3	Dimensionality Reduction and Visualization	2
3	Analysis of Feature Extraction Power	3
3.1	Before Transfer Learning: A Poor Fit	3
3.2	After Transfer Learning: An Adapted Powerhouse	3
4	Visualization Using Dimensionality Reduction	3
4.1	Principal Component Analysis (PCA)	4
4.2	t-Distributed Stochastic Neighbor Embedding (t-SNE)	4
4.3	Uniform Manifold Approximation and Projection (UMAP)	5
5	Conclusion	6

1 Introduction

Convolutional Neural Networks (CNNs) pre-trained on large-scale datasets like ImageNet have demonstrated remarkable capabilities in learning rich, hierarchical feature representations from images. This learned knowledge, often referred to as a model’s "feature extraction power," can be transferred to new, different tasks through a process called transfer learning.

This report investigates and visualizes this concept using the **DenseNet121** model, pre-trained on ImageNet. The core objective is to analyze the nature of its extracted features when applied to a distinctly different dataset—the MNIST database of handwritten digits.

We will explore two scenarios:

1. **Before Transfer Learning:** Using the DenseNet121 model as a "zero-shot" feature extractor for MNIST digits.
2. **After Transfer Learning:** After fine-tuning the DenseNet121 model on the MNIST dataset.

To analyze the high-dimensional feature vectors produced by the model in each scenario, we will use three powerful dimensionality reduction techniques: **Principal Component Analysis (PCA)**, **t-Distributed Stochastic Neighbor Embedding (t-SNE)**, and **Uniform Manifold Approximation and Projection (UMAP)**. By projecting the features onto a 2D plane, we can visually assess their separability and, by extension, the model’s adapted understanding of the new dataset.

2 Methodology

The experiment follows a systematic process to extract, reduce, and visualize feature vectors.

2.1 Model and Datasets

- **Pre-trained Model: DenseNet121**, with weights trained on the ImageNet dataset. The model’s convolutional base is used for feature extraction.
- **Target Dataset: The MNIST dataset**, which consists of 70,000 grayscale images of handwritten digits (0-9), each of size 28×28 . For use with DenseNet121, these images were converted to RGB (by duplicating the single channel three times) and resized to 48×48 pixels to meet the model’s input requirements.

2.2 Feature Extraction Process

- **Before Transfer Learning:** MNIST test images are passed through the frozen convolutional base of the original ImageNet-trained DenseNet121. The output from the final pooling layer is a high-dimensional feature vector for each image.
- **After Transfer Learning:** A new classification head (e.g., a Flatten layer followed by Dense layers) is added to the DenseNet121 base. The model is then fine-tuned on the MNIST training set. After training, the fine-tuned convolutional base is used to extract feature vectors from the MNIST test images.

2.3 Dimensionality Reduction and Visualization

The extracted feature vectors are high-dimensional (e.g., 1024 dimensions for DenseNet121). To visualize them, we use the following techniques to project them into 2D space:

- **PCA:** A linear technique that projects data onto the dimensions of highest variance.

- **t-SNE:** A non-linear technique that excels at revealing the local clustering structure of data.
- **UMAP:** A non-linear technique known for its speed and its ability to preserve both local and global data structure.

The resulting 2D points are plotted and color-coded according to their true digit class (0-9).

3 Analysis of Feature Extraction Power

The quality of the extracted features is determined by how well they separate the different classes of digits.

3.1 Before Transfer Learning: A Poor Fit

When the original DenseNet121 model is used to extract features from MNIST digits, its "power" is mismatched for the task. The model was trained to find features relevant to real-world objects in ImageNet (e.g., textures of fur, shapes of cars, patterns on dogs). These features are not inherently useful for distinguishing between handwritten digits.

As a result, the extracted feature vectors for different digits are expected to be very similar. When projected into 2D space, the points corresponding to the 10 digit classes (0-9) will likely be heavily overlapped and jumbled together in a single, large cloud. This demonstrates that the model's original feature extraction capability is not generalizable to this new, dissimilar domain.

3.2 After Transfer Learning: An Adapted Powerhouse

Through fine-tuning, the model adapts its feature extraction power. It learns to "see" features relevant to digits, such as loops (for 8s and 9s), straight lines (for 1s and 7s), and intersections. The weights in its deeper convolutional layers are updated to recognize these new patterns.

Consequently, the feature vectors extracted from the fine-tuned model should be highly discriminative. When these vectors are projected into 2D, we expect to see a dramatic improvement in class separation. The plot should reveal **ten distinct, well-separated clusters**, where each cluster corresponds to one digit class. This clear separation visually confirms that transfer learning has successfully retrained the model's feature extraction power to master the new dataset.

4 Visualization Using Dimensionality Reduction

The following plots visualize the high-dimensional feature vectors on a 2D plane.

4.1 Principal Component Analysis (PCA)

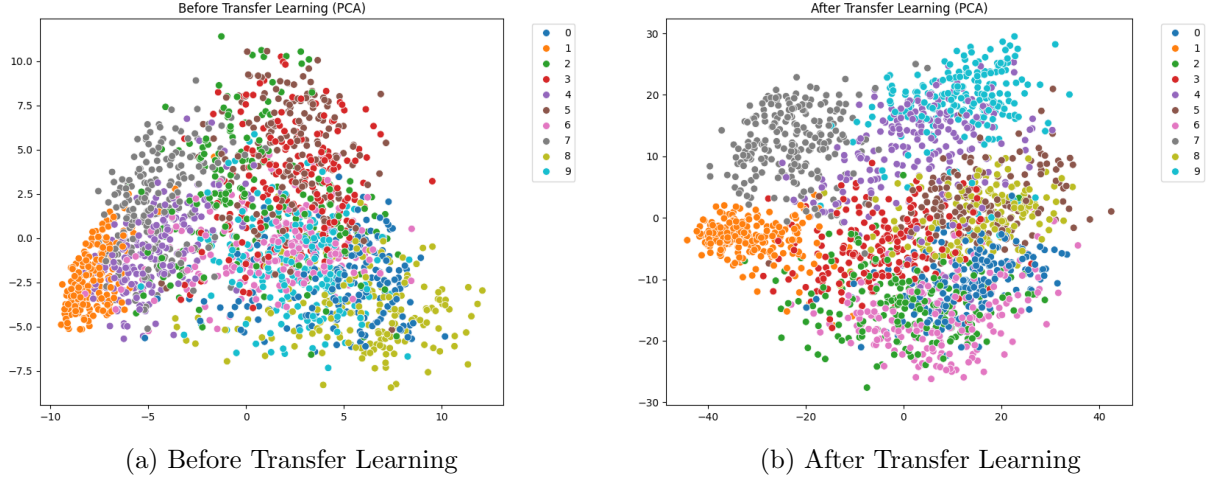


Figure 1: PCA visualization of MNIST features. Before fine-tuning, the classes are poorly separated. After fine-tuning, distinct clusters begin to form.

4.2 t-Distributed Stochastic Neighbor Embedding (t-SNE)

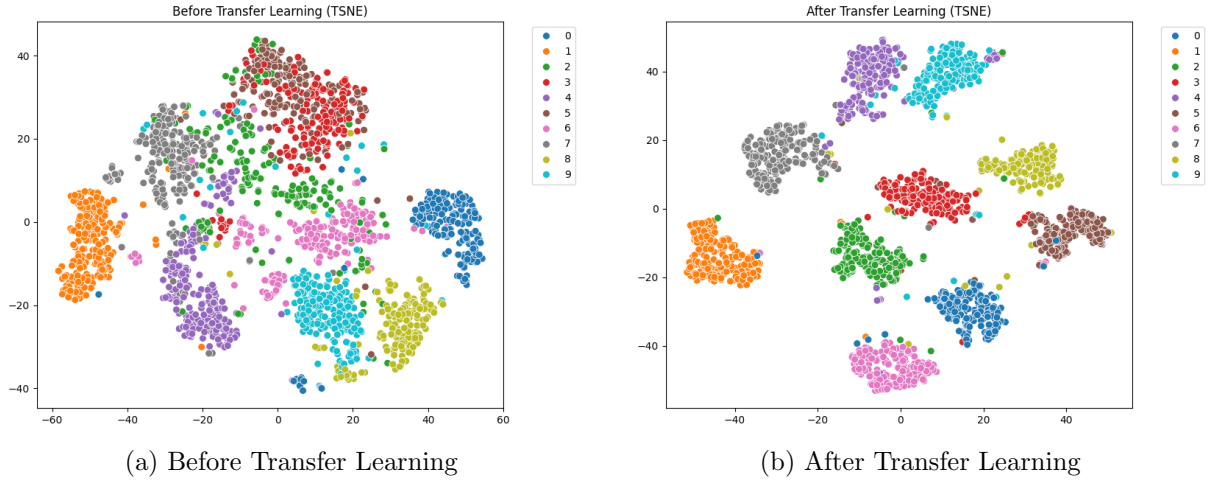


Figure 2: t-SNE visualization of MNIST features. After fine-tuning, t-SNE reveals extremely tight and well-separated clusters, highlighting its strength in visualizing local structure.

4.3 Uniform Manifold Approximation and Projection (UMAP)

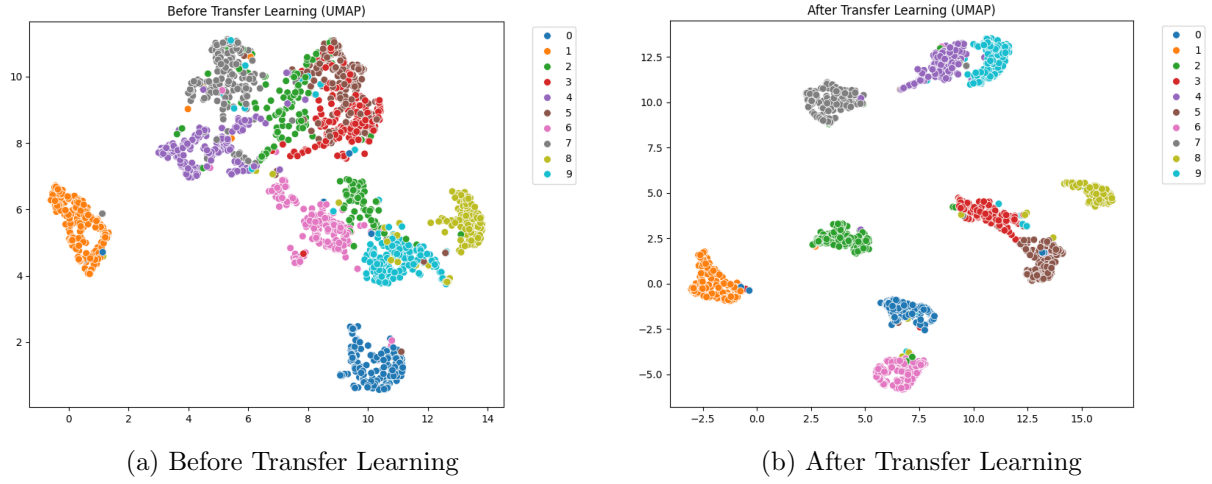


Figure 3: UMAP visualization of MNIST features. Similar to t-SNE, UMAP shows clear, dense clusters after transfer learning, often with better preservation of the global relationships between clusters.

5 Conclusion

This report successfully visualized the impact of transfer learning on the feature extraction power of a pre-trained CNN. The visualizations clearly demonstrate that:

- **Before fine-tuning**, the feature space learned from ImageNet is not suitable for the MNIST digit recognition task, resulting in poor class separability.
- **After fine-tuning**, the model adapts its internal representations, creating a new feature space where each digit class forms a distinct and separable cluster.

Both t-SNE and UMAP proved to be superior to PCA for visualizing the non-linear structures in the feature space, revealing clean, tight clusters after training. This investigation confirms the fundamental principle of transfer learning: pre-trained models are not just static feature extractors but can be effectively repurposed and adapted to master entirely new domains, a process made strikingly clear through visualization.

Assignment Github Link : Assignment